MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS)

END SEMESTER EXAMINATION

SEMESTER IV

DSE – MG4DSEMATA00 – INTRODUCTION TO MACHINE LEARNING

(2024 Admission onwards)

MODEL QUESTION PAPER

Duration: 2 Hours **Maximum Marks: 70**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C)

Students should attempt at least one question from each course outcome.

Part A

One or two sentences

Answer any **5** questions

Each question carries **2** marks

1. What is machine learning? [U/CO:1]
2. Differentiate: categorical versus numeric attribute [U/CO:2]
3. Define a model. [E/CO:3]
4. Why do we need to calculate Kappa value for a classification model? [E/CO:3]
5. What is the Hamming distance between binary vectors 1001 and 0101? [U/CO:4]
6. What is a feature? Explain with an example. [A/CO:4]
7. Define slope in a linear regression. [A/CO:5]
8. What is heteroskedasticity? [A/CO:5]

[2 × 5 = 10]

Part B

Short answer

Answer any **5** questions

Each question carries **6** marks

9. Explain the process of abstraction. [U/CO:1]

10. Explain qualitative and quantitative data in details. [U/CO:1]

11. Explain "No Free Lunch" theorem in context of machine learning. [E/CO:3]

12. Differentiate between predictive and descriptive models. [E/CO:3]

13. When can a feature be termed as irrelevant? How can it be measured? [A/CO:4]

14. Differentiate feature transformation with feature selection [A/CO:4]

15. Explain logistic regression. [A/CO:5]

16. Explain maximum likelihood estimation. [A/CO:5]

[6 × 5 = 30]

Part C

Essays
Answer any **3** questions
Each question carries **10** marks

17. Compare the different forms of machine learning with those of human learning. [U/CO:1]

18. (a) Discuss an application of machine learning in healthcare.
    (b) Elucidate association analysis.

    [U/CO:1]

19. What are the main activities involved in machine learning? What is meant by data pre-processing? [U/CO:1]

20. Explain in detail
    (a) Holdout method
    (b) 10-fold cross-validation method of training a model.

    [E/CO:3]

21. (a) Explain the overall process of feature selection
    (b) Explain, with an example, the main underlying concept of feature extraction. What are the most popular algorithms for feature extraction?

    [U/CO:4]

22. From the following data, find the regression equation of Y on X:

X	15	23	18	23	24	22	22	19	19
Y	49	63	58	60	58	61	60	63	60

[A/CO:5]

[10 × 3 = 30]